$$\frac{20\,km}{5\,hours}=4\,km\ per\ hour$$

where the units of the first term is a km and the unit of the second term is an hour or reciprocally,

$$\frac{5\,hours}{20\,km}=\frac{1}{4}\ hour\ per\ km\ .$$

where the unit of the first term is an hour and the unit of the second term is a km.

Example 8.12

An automobile driver travels from plain to hill station 100km distance at an average speed of 30km per hour. He then makes the return trip at average speed of 20km per hour what is his average speed over the entire distance (200km)?

Solution:

If the problem is given to a layman he is most likely to compute the arithmetic mean of two speeds.

i.e., $\overline{X} = \frac{30\,km + 20\,km}{2} = 25\ kmph$

But this is not the correct average.

So harmonic mean would be mean suitable in this situation.

Harmonic Mean of 30 and 20 is

$$HM=\frac{2}{\left(\frac{1}{20}\right)+\left(\frac{1}{30}\right)}=\frac{2}{\left(\frac{5}{60}\right)}=\frac{2(60)}{5}=24\,kmph$$

8.1.6 Relationship among the averages

In any distribution when the original items differ in size, the values of AM, GM and HM would also differ and will be in the following order

$$AM \geq GM \geq HM$$



If all the numbers $X_1, X_2, \ldots, X_n$ are identical then, AM = GM = HM.

Example 8.13

Verify the relationship among AM, GM and HM for the following data

X	7	10	13	16	19	22	25	28
f	10	22	24	28	19	9	12	16

Solution:

X	f	Xf	log$X$	f log$X$	f / X
7	10	70	0.8451	8.4510	1.4286
10	22	220	1	22.0000	2.2000
13	24	312	1.1139	26.7346	1.8462
16	28	448	1.2041	33.7154	1.7500
19	19	361	1.2788	24.2963	1.0000
22	9	198	1.3424	12.0818	0.4091
25	12	300	1.3979	16.7753	0.4800
28	16	448	1.4472	23.1545	0.5714
	$\sum f = N = 140$	$\sum fX = 2357$		$\sum f \log x = 167.209$	$\sum \frac{f}{x} = 9.6852$

Table : 8.10

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